AEROLINK X7

Smart Communication Hub • Product Specification & Technical Data Sheet

Field	Specification
Commercial name	AeroLink X7 Smart Communication Hub
Model	ALX7-X7-240
Manufacturer	Northstar Connected Devices Ltd.
Product family	AeroLink X Series
Product type	Networked communication and control appliance
Country of manufacture	Fictional Republic of Norland
Primary function	Aggregates wired/wireless communication interfaces and coordinates connected devices
Typical installation	Indoor fixed installation; desktop or wall-mounted

1. Product Overview

The AeroLink X7 is a compact communication hub designed to collect information from connected peripherals and make that information available to authorized local applications. The unit combines a processing board, network interfaces, wireless modules, power conversion circuitry, storage, status indicators, and a protective enclosure.

The device is supplied as a complete functional unit. It is not marketed as a general-purpose computer. Its software is pre-installed and is used primarily for communication management, device coordination, diagnostics, and local configuration.

The X7 can operate as a standalone hub or as part of a larger facility network. Depending on installation, it can communicate with barcode scanners, environmental sensors, access-control readers, handheld terminals, and industrial gateways.

2. Functional Architecture

Subsystem	Description	Primary role
Main processor board	Quad-core embedded processing board	Local processing and protocol management
Ethernet subsystem	Two 1 GbE copper interfaces	Wired network communication
Wireless subsystem	Wi-Fi 6 and Bluetooth Low Energy	Local wireless communication
Storage	128 GB solid-state flash	Software, configuration and logs
Power subsystem	External DC input with internal regulation	Internal power rails
Indicator board	Status LEDs and service button	Operational status and diagnostics
Expansion	Two USB-A and one service connector	Peripheral and maintenance connectivity

3. Detailed Technical Specifications

Parameter	Value	Notes
Processor	4-core ARM-based embedded processor, 1.8 GHz	Soldered
Memory	4 GB LPDDR4	Non-removable
Storage	128 GB eMMC/flash	Firmware and logs
Ethernet	2 × 10/100/1000 Mbps RJ45	Auto-negotiation
Wi-Fi	802.11 a/b/g/n/ac/ax	2.4/5 GHz
Bluetooth	Bluetooth Low Energy	Peripheral communication
USB	2 × USB-A 3.0; 1 × USB-C service	Peripheral/service
Power input	12 V DC, 3 A maximum	External adapter
Consumption	18 W typical	Varies
Operating temperature	0 °C to 45 °C	Indoor
Dimensions	178 × 118 × 36 mm	Excluding bracket
Weight	0.74 kg	Main unit
Enclosure	PC/ABS polymer with metal shielding	Molded enclosure

4. Communications and Interfaces

Ethernet is intended for the primary facility connection. Wi-Fi provides an alternative network path or local wireless connectivity, while Bluetooth Low Energy is used for short-range peripherals. USB ports support selected accessories and service tools. The wireless modules are integrated into the main electronics assembly and are not intended to be removed during ordinary installation.

5. Electrical Characteristics

Item	Specification
Input connector	5.5 mm DC barrel-type connector
Nominal input	12 V DC
Maximum input current	3 A
Protection	Over-voltage, over-current and thermal protection
External adapter	12 V DC / 36 W adapter supplied with selected kits
Battery	No internal battery

6. Mechanical Construction

The housing consists primarily of molded PC/ABS polymer panels surrounding a populated printed circuit board assembly. A small internal metal shield is fitted over the radio and processor area. The enclosure includes mounting bosses, ventilation openings, LED light pipes, and a removable service cover.

Material/component	Role
PC/ABS enclosure	Mechanical protection and mounting
Internal metal shield	Electromagnetic shielding
Printed circuit board	Processor, memory, interfaces and power circuits
Aluminum mounting bracket	Optional mounting
Thermal interface materials	Heat transfer
Fasteners	Assembly

7. Included Items and Packaging

Package item	Qty	Standard unit?
AeroLink X7 main unit	1	Yes
12 V DC power adapter	1	Yes
Ethernet cable, 1 m	1	Yes
Wall mounting bracket	1	Yes
Mounting screw set	1 set	Yes
Quick installation guide	1	Yes
USB service cable	1	No, service kit

8. Product Variants

The X7 family is sold in several configurations. The ALX7-X7-240 configuration includes dual Ethernet, Wi-Fi 6, Bluetooth Low Energy, 4 GB memory, and 128 GB storage. A visually similar ALX7-X7-120 variant has reduced storage and a separate product code. The two variants share substantially similar enclosures.

9. Manufacturing and Assembly

The main electronics assembly is populated by automated surface-mount manufacturing. Processor, memory, storage, radio modules, controllers, power-management components, connectors, and passive components are mounted on a multi-layer printed circuit board. The populated board is electrically tested before enclosure installation. Final assembly includes board installation, cable routing, shielding, functional testing, labeling, and packaging.

10. Software and Intended Use

The X7 is delivered with embedded firmware and a local management interface. The software handles network configuration, connected-device communication, diagnostics, event logging, and authorized application interfaces. The primary intended use is communication and coordination between network-connected devices within a facility. It is not designed for general office productivity.

11. Identification and Packaging Notes

Label location	Information shown
Product underside	Model, serial number, input rating and country-of-origin statement
Outer carton	Model, package contents, handling symbols and fictional barcode
Main board	Assembly revision and internal board identifier
Power adapter	Input/output rating and adapter model

12. Technical Notes

Note A: Some interface specifications vary by firmware revision. Note B: Weight excludes optional mounting hardware. Note C: Wireless modules are integrated into the finished appliance. Note D: Packaging configuration may vary by customer order.

End of Product Specification